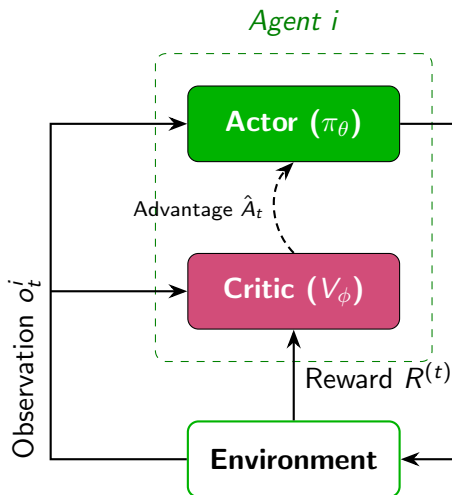


Theoretical Background: Actor-Critic Architecture



► **Actor (π_θ):** Outputs the policy—deciding which cognitive action $a_t^i \in \{\text{Retrieve, Disseminate, Critique}\}$ to take based on the local observation o_t^i .

► **Critic (V_ϕ):** Evaluates the current state by predicting the expected return $V_\phi(o_t^i)$ to guide the Actor's updates.

► **Environment:**

Represents the collaborative writing process (maintaining D, I, G, E). It transitions